

Software Engineering Lab 2

Agile Backlog Creation & Sprint Simulation in Jira

Remote Patient Vitals Alert & Monitoring App

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This report converts the Lab 1 functional requirements into Jira-style epics and user stories, prioritizes and estimates them, and documents a one-week Sprint 1 simulation.

1. Epics

Epic	Description	Lab 1 coverage
EPIC-1: Continuous Vital Telemetry	Reliable collection and viewing of remote patient vital readings.	FR-001, FR-005
EPIC-2: Threshold Detection and Alerts	Detect abnormal readings and provide actionable alert information.	FR-002, FR-003
EPIC-3: Caregiver Notification and Escalation	Deliver alerts to the correct caregiver and escalate emergencies.	FR-004
EPIC-4: Alert Response and Audit	Record caregiver responses and preserve traceable event history.	FR-005

2. User Story Backlog

ID	Epic	User story	Priority	Points
US-001	EPIC-1	As a Remote Patient, I want my registered device to send timestamped SpO2, heart-rate, and blood-pressure readings so that my caregiver can monitor me remotely.	High	5
US-002	EPIC-1	As an On-Call Caregiver, I want to view current and recent patient vitals so that I can assess the patient's condition.	Medium	3
US-003	EPIC-2	As an On-Call Caregiver, I want every valid reading evaluated against active clinical thresholds so that anomalies are detected consistently.	High	5
US-004	EPIC-2	As an On-Call Caregiver, I want an alert to show the metric, value, threshold, patient, and time so that I can respond quickly.	High	5
US-005	EPIC-3	As an assigned On-Call Caregiver, I want an immediate critical-alert notification so that I can begin a response.	High	3
US-006	EPIC-3	As a care-team member, I want an unacknowledged critical alert escalated through the caregiver matrix so that emergencies are not missed.	High	5
US-007	EPIC-4	As an On-Call Caregiver, I want to acknowledge an alert and record an action or note so that the response is documented.	Medium	3
US-008	EPIC-4	As an On-Call Caregiver, I want notification attempts and acknowledgements timestamped so that the event history is traceable.	Low	3

3. Sprint 1 Simulation

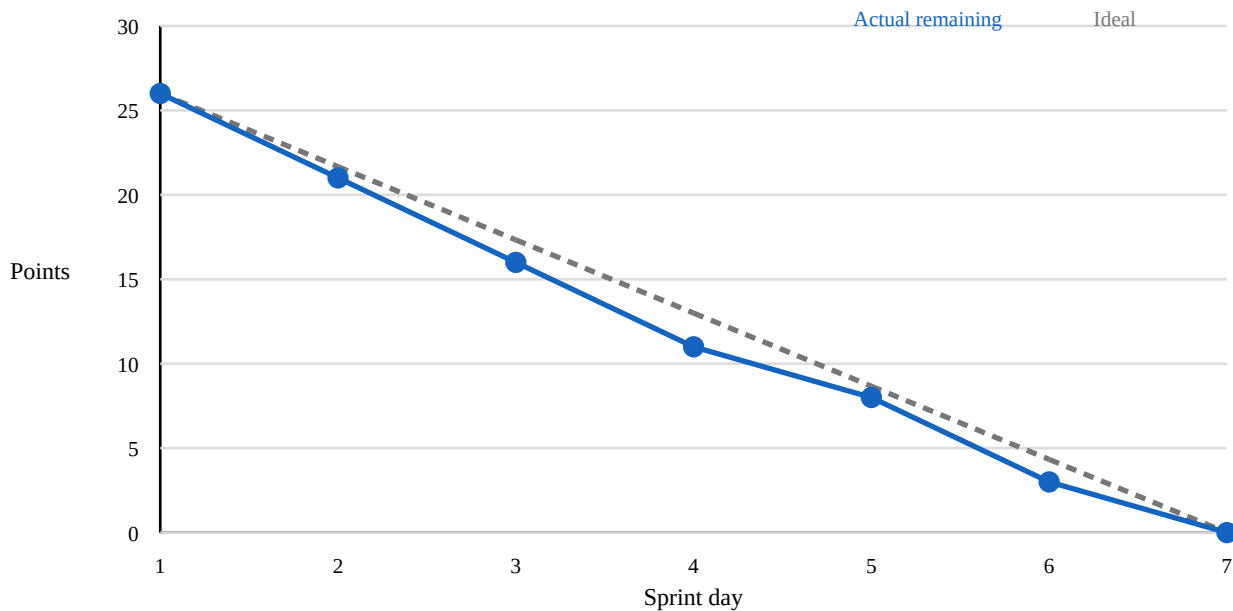
Sprint duration: one week. Selected stories: US-001, US-003, US-004, US-005, US-006, and US-007. Total commitment: **26 story points**. US-002 and US-008 remain in the backlog.

Day	Completed work	Remaining points
Day 1	Sprint started; selected stories in To Do	26
Day 2	US-001	21
Day 3	US-003	16
Day 4	US-004	11
Day 5	US-005	8
Day 6	US-006	3
Day 7	US-007; Sprint completed	0

4. Burndown Chart

The red/blue actual line represents remaining story points as work moves from To Do through In Progress to Done. The dashed line is the ideal linear guideline.

Sprint 1 Burndown Chart



5. Reflection

Did the estimates reflect the actual effort?

The estimates were reasonably aligned with the simulated effort. Three-point stories covered focused notification or response actions, while five-point stories involved integration, clinical rules, or escalation behavior. Real device integration and security testing could increase the effort.

Was the backlog well-prioritized?

Yes. The high-priority path establishes monitoring, detection, alert generation, notification, and escalation before lower-priority viewing and audit improvements. This keeps emergency safety work ahead of convenience features.

How did the simulated sprint align with your plan?

The sprint completed all six selected stories and delivered 26 points within the one-week simulation. The order followed the dependency chain from telemetry to detection, notification, escalation, and caregiver response.

What insights did the burndown chart give about team capacity?

The chart reaches zero by Day 7, indicating that the simulated team could complete the 26-point commitment. The two unselected stories show why future sprint commitments should remain within demonstrated capacity. A real Jira chart would also expose delays and scope changes.

6. Jira Evidence Checklist

- Backlog screenshot showing all four epics and eight user stories.
- Story point and priority fields visible for the stories.
- Active Sprint screenshot showing work in To Do, In Progress, and Done.
- Reports > Burndown Chart screenshot.
- This PDF uploaded to the GitHub repository.